	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence 3** **Statistical models and research planning** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Name: _____ **Date:** _____

Answer on separate paper and show your reasoning. Use the supplied normal and Student's t reference sheets. Report t critical values and normal critical values from the ∞ row to **three decimal places**, and normal probabilities to **four decimal places**. For t cutoffs between printed entries, give probability bounds without interpolation. Each criterion is worth **6 points** (**18 points** total); at least **4 points per criterion** are required to pass that criterion.

Questions 1–6 concern independent random observations from normal populations. Here n is sample size, σ is population standard deviation, s is observed sample standard deviation, and ν denotes degrees of freedom. T (Questions 1–3) and W (Questions 4–6) denote the standardized sample mean relative to the population mean; cutoffs are dimensionless.

1. Export inspection costs [C8-2]

An exporter models the standardized mean inspection cost from $n = 12$ shipments. Population σ is unknown and estimated from the sample.

(a) Determine ν and find $a > 0$ with $P(T > a) = 0.01$. **[1]**

(b) Find $b < 0$ with $P(T < b) = 0.05$. Justify the sign and table column. **[1]**

2. Market-stall electricity use [C8-2]

A study models the standardized mean electricity use from $n = 25$ stalls, estimating unknown population σ from the sample.

(a) Determine ν and convert a central area of 0.98 into the combined outside area and the area in each tail. **[1]**

(b) Find both symmetric cutoffs enclosing this central area. Identify the table column used. **[1]**

3. Cold-storage energy costs [C8-2]

The standardized mean energy cost for $n = 15$ facilities uses a sample estimate of unknown population σ.

(a) Determine ν and give bounds for $P(T > 2.40)$. **[1]**

(b) Give bounds for both $P(T < -2.40)$ and $P(-2.40 < T < 2.40)$. **[1]**

4. Bottling-line measurements [C9-2]

A sample contains $n = 16$ bottle volumes. The population standard deviation is known to be $\sigma = 3$ mL.

(a) Choose and justify the model for W. Find the negative cutoff c with $P(W < c) = 0.025$. **[1]**

(b) Calculate $P(-1.00 < W < 1.50)$. **[1]**

5. Repair-time averages [C9-2]

A workshop samples $n = 10$ repairs. Population σ is unknown; the sample gives $s = 4$ minutes.

(a) Choose and justify the model, determine ν, and find the positive cutoff exceeded with probability 0.025. **[1]**

(b) Estimate $P(-1.833 < W < 2.821)$. **[1]**

6. Comparing parcel-mass studies [C9-2]

Study A uses $n = 12$ parcels and known population $\sigma = 0.4$ kg. Study B uses $n = 12$ parcels and estimates unknown σ using $s = 0.4$ kg.

(a) Justify each model and find its symmetric cutoffs enclosing 0.95. Calculate the difference between positive cutoffs and explain it using tail behaviour. **[1]**

(b) If B increases to $n = 61$ while σ remains unknown, give its model, degrees of freedom, and new 0.95 central cutoffs. Explain the trend toward the normal model and why sample size alone does not determine the choice. **[1]**

7. Plan your future research [C10-6]

Plan three different real-life situations using actual observations: a continuous variable with a linear density on a stated interval; exponential waiting times or intervals between events; and normal measurements with a population-mean investigation. Address all three situations in each part. Submit a research plan only; no collected data, completed calculations, working links, or finished article are required.

(a) For each situation, specify the context, variable, units, population, time period, and research objective. For the normal situation, state a specific population-mean question. **[2]**

(b) For each situation, identify a real public dataset and source, or give a feasible search strategy specifying a public source, search terms, variables, and observation coverage. Describe planned data selection, organization, missing-value checks, and filters. Plan a

histogram with interval choices, sample mean, variance, and standard deviation. How will you compare these with theoretical model statistics, obtain the required parameters, and address any missing information? For normal measurements, distinguish the distribution comparison from mean inference. **[2]**

(c) Justify each proposed model and state the assumptions and evidence you will check. For the normal situation, justify the planned normal or Student's t model for mean inference, addressing sampling, independence, and whether population σ is known or estimated; specify how degrees of freedom will be determined if applicable. Identify one anticipated limitation or model mismatch for each situation. Why are histograms alone insufficient to establish model suitability, and why must your choices remain provisional? **[2]**